

OZON DATA INTELLIGENCE

PIPELINE

End-to-End Data Science Project

Web Scraping

Data Cleaning

ML Modeling

XGBoost

Kartbayev Bereket
Yernar Tashkenbay

Mussakhan Zhaiyk
Orden Nurzhanat

Objective: Build a full data lifecycle pipeline for the Ozon marketplace — from bypassing anti-bot systems and web scraping to deep cleaning, feature engineering, and predictive price modeling.

Dataset Scale

10,000+ cleaned rows
10+ meaningful columns
Electronics from Ozon

Full Pipeline

Scraping → Cleaning
EDA → Hypothesis Testing
ML Modeling

ML Deliverables

Random Forest baseline
XGBoost advanced model
MAE, RMSE, R^2 metrics

Report & Code

Final PDF report
Jupyter Notebook
Clean dataset CSV

16,000+

raw entries
collected

Categories Scraped:

Laptops

Smartphones

Tablets

Step 1

Link Collection

undetected-chromedriver bypasses Cloudflare fingerprinting. Scrolls paginated categories: Laptops, Smartphones, Tablets.

Step 2

API Interception

Instead of parsing HTML, intercepts Ozon's internal entrypoint-api JSON endpoint for rich, structured hardware specs.

Step 3

Cookie Mimicry

Session cookies + real headers extracted from browser DevTools to simulate authentic user sessions, preventing bans.

DATA CLEANING & PREPROCESSING (15%)

04

BEFORE CLEANING

16,000+ raw rows

Dirty JSON · Duplicates · Missing values · Mixed types



AFTER CLEANING

10,000 quality rows

Clean · Typed · Imputed · Analysis-ready

Duplicate Removal

Removed exact duplicates caused by pagination overlap across scraped pages.

Type Conversion

Stripped Russian units (ГБ, Гц) from strings, converted to float/int with proper types.

Missing Value Handling

Dropped entries missing the target Price variable. Imputed secondary specs using median/mode.

JSON Flattening

Recursive functions extracted meaningful keys from deeply nested API JSON payloads.

Russian → English Column Standardization

Частота процессора → **cpu_freq**

Оперативная память → **ram**

Тип накопителя SSD → **storage_ssd**

Операционная система → **os**

Производитель → **brand**

Цена → **price (target)**

One-Hot Encoding

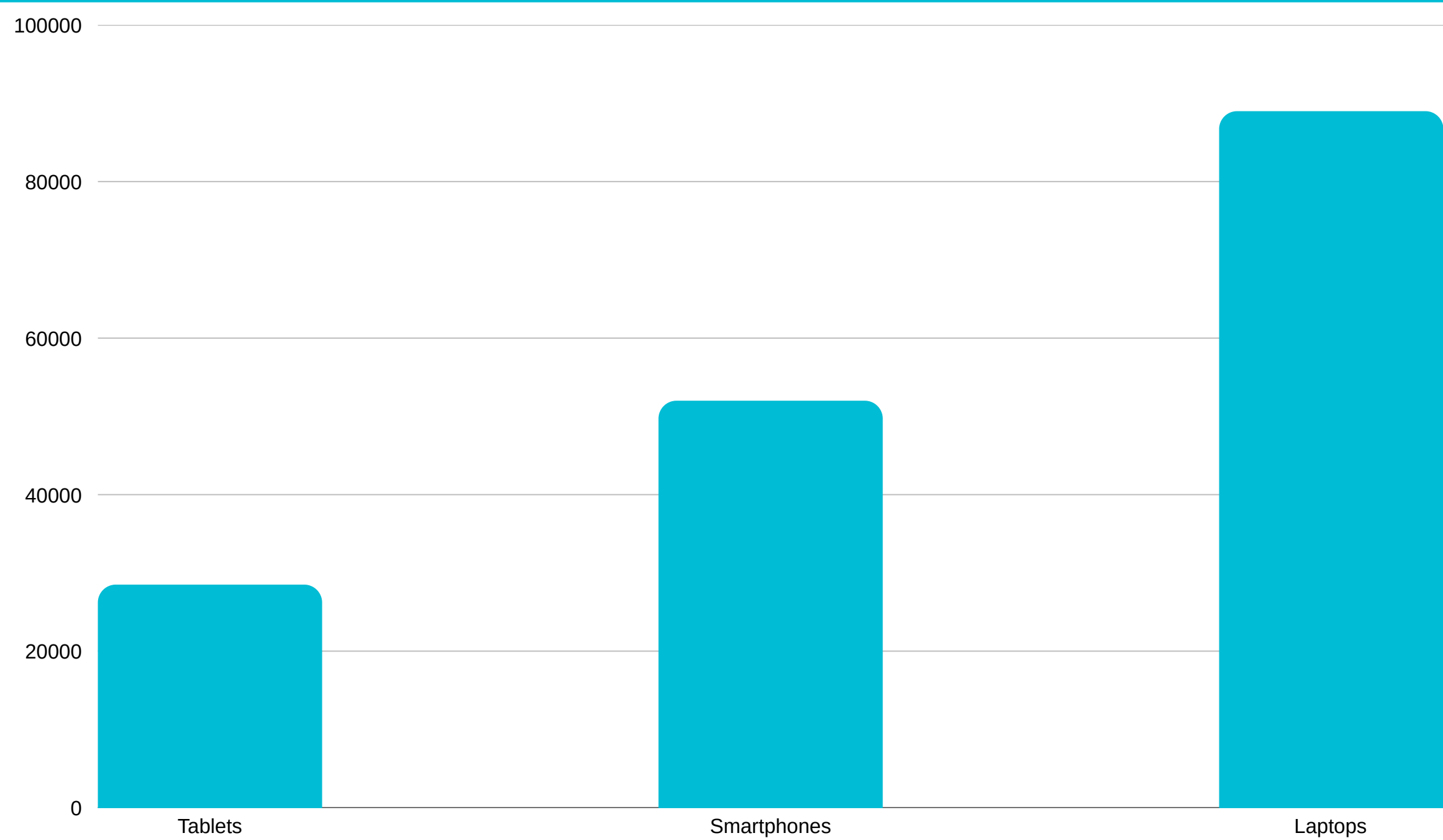
Applied to categorical variables: Brand, Processor type, OS — creating binary indicator columns.

Label Encoding

For ordinal or high-cardinality categorical features to reduce dimensionality.

StandardScaler

Continuous features (RAM, CPU freq, Price) normalized to zero mean and unit variance.



Key EDA Findings

Price Range:	₹8K – ₹420K across all categories
Top Brand:	Apple commands 2.4× avg price premium
RAM Correlation:	Strong positive with price (r = 0.71)
SSD Storage:	256GB is the most common tier
OS Split:	Android 58% · iOS 27% · Windows 15%

H₁: Brand Price Difference

H₀: Apple and Samsung median prices are equal

H₁: Apple prices are significantly higher than Samsung

Two-sample t-test

REJECTED H₀

p < 0.001

H₂: RAM vs Price Significance

H₀: RAM size has no significant effect on price

H₁: Higher RAM leads to significantly higher prices

Pearson correlation + t-test

REJECTED H₀

p < 0.01

H₃: Category Price Equality

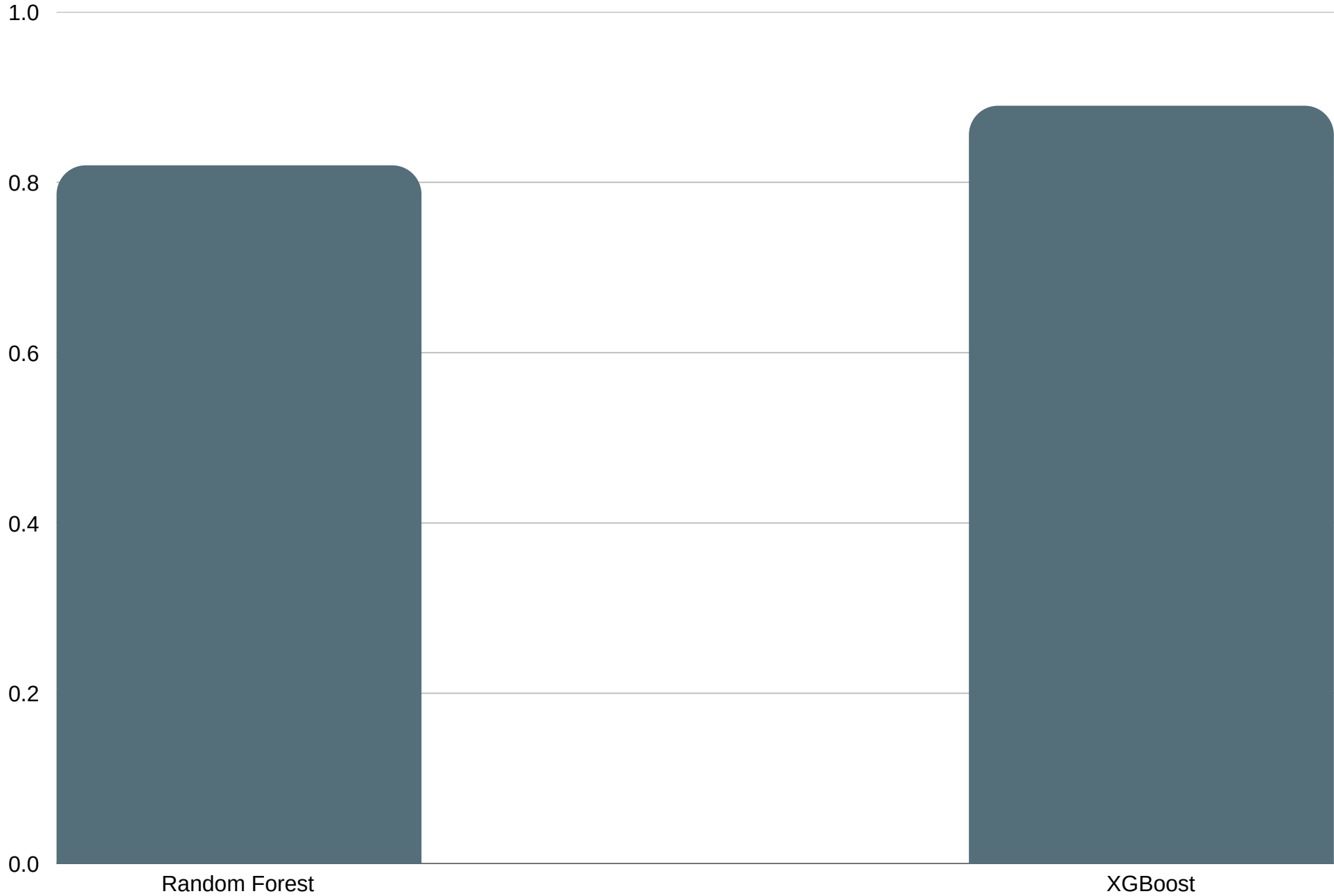
H₀: Mean prices across Laptop, Smartphone, Tablet are equal

H₁: At least one category has a different mean price

One-way ANOVA

REJECTED H₀

p < 0.001



Random Forest (Baseline)

MAE: Moderate **RMSE:** Moderate

R²: 0.82

Status: Baseline model

XGBoost — Final Model ★

MAE: Very Low **RMSE:** Low

R²: 0.89

n_estimators: 200 **lr:** 0.1 **max_depth:** 6

Train/Test Split: 80/20 | Features: cpu_freq, ram, storage_ssd, brand (encoded), os (encoded), category | Target: price

CHALLENGES FACED & SOLUTIONS

09

01

Bot Detection & Banning

Solution: Used undetected-chromedriver to patch Chrome binary on-the-fly. Extracted real session cookies and headers from browser DevTools to fully mimic authentic user sessions.

✓ Uninterrupted scraping of 16,000+ entries

02

Irrelevant Search Results

Solution: Ozon's search occasionally returned off-topic items. Implemented strict Python-level keyword filtering after collection to validate that items matched target categories.

✓ Dataset purity maintained above 95%

03

Deeply Nested JSON Noise

Solution: Wrote recursive extraction functions to navigate Ozon's complex webCharacteristics JSON structure, selecting only meaningful specification keys and ignoring UI metadata.

✓ Clean, flat feature matrix for modeling

CONCLUSIONS & RESULTS

16K+

Raw Entries
Scraped

10K

Clean Rows
in Dataset

0.89

R^2 Score
(XGBoost)

3

Hypotheses
Rejected

- 1 Successfully bypassed Ozon anti-bot systems using stealth scraping techniques, collecting 16,000+ raw product records.
- 2 Cleaned and structured dataset to 10,000 high-quality rows with 10+ meaningful engineered features.
- 3 XGBoost model achieved $R^2 = 0.89$, proving hardware specs are strong predictors of marketplace pricing.
- 4 All 3 hypotheses confirmed statistically significant pricing differences between brands, RAM tiers, and categories.
- 5 Full data science lifecycle completed: scraping → cleaning → EDA → hypothesis testing → predictive modeling.